

PARTIAL DIFFERENTIAL EQUATIONS: A COMPILATION

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In these notes we shall state, mostly without proof, important results from elementary theory of partial differential equations in $\mathbb{R}^n$. We focus on the “classical problems”, i.e. transport equations, wave & heat equations, and properties of harmonic functions. We cover the basics of distributions and weak solutions. We briefly explore important results from the theory of Fourier series.

These notes are not meant as a sole reference, in fact, these would serve as poor course notes as they do not contain proofs. These notes are meant as more of a *crash course* for students enrolled in Math 475.

1. TRANSPORT EQUATIONS

Throughout this section, we consider some transport equations and illustrate the method of characteristics. The method of characteristics cannot be elegantly stated, but it can be demonstrated. This will make the first section of our text quite different from those that will appear later. In fact, this section will comprise mostly of one or two non-trivial (but non-tedious) examples of the method of characteristics.

Example 1. Find a smooth solution to the following PDE

$$u_{x_1} + x_1 u_{x_2} + u_{x_3} \equiv u, \quad u(x_1, x_2, 0) = x_1 + x_2.$$

Solution. Here we represent the spatial variables by $(x_1, x_2, x_3) \in \mathbb{R}^3$. The PDE gives us information about the solutions along specific curves in space. Define

$$\dot{x}_1(s) := 1, \quad \dot{x}_2(s) = x_1(s), \quad \dot{x}_3(s) = 1.$$

Suppose we can solve for the parametrization of such curves, i.e. obtain functions $x_j(s)$ satisfying the above (s is some real dummy variable). Let $z(s)$ be the value of a solution along these curves:

$$z(s) := u(x_1(s), x_2(s), x_3(s)).$$

Then an easy calculation gives

$$\dot{z}(s) = z(s)$$

so that $z(s) = Ce^s$ for some $C \in \mathbb{R}$. Direct observation can give solutions

$$x_1(s) = s + c_1, \quad x_2(s) := \frac{s^2}{2} + c_1s + c_2, \quad x_3(s) = s$$

to our system of ODEs. Here we have chosen $x_3(s)$ such that $x_3(0) = 0$. Then, the initial conditions give us

$$x_1(0) + x_2(0) = u(x_1(0), x_2(0), 0) = z(0) = C.$$

Hence, $u(x_1, x_2, x_3) = (c_1 + c_2)e^{x_3}$. Clearly, $c_1 = x_1 - s = x_1 - x_3$. This allows us to write

$$x_2(s) = x_1(s)x_3(s) - \frac{x_3(s)^2}{2} + c_2$$

whence

$$u(x_1, x_2, x_3) = \left[x_1 - x_3 + x_2 + \frac{x_3^2}{2} - x_1x_3 \right] e^{x_3}.$$

□

Example 2. Using the method of characteristics, find a solution to

$$u_t + uu_x \equiv 1, \quad u(x, 0) = x, \quad (x, t) \in \mathbb{R} \times (0, \infty).$$

Solution. Here our characteristics are given by

$$\dot{t}(s) = 1, \quad \dot{x}(s) = z(s)$$

where $z(s) = u(x(s), t(s))$. A direct calculation gives $\dot{z}(s) = 1$ so that $z(s) = s + c$. To solve our characteristics, we consider

$$t(s) = s, \quad x(s) = \frac{s^2}{2} + cs + c'.$$

Obviously, we have chosen the parametrization of t so that $t(0) = 0$. Thus,

$$u(x(0), t(0)) = z(0) = c = x(0) = c'.$$

Hence,

$$x(s) = \frac{t^2}{2} + c(t + 1).$$

Combing these facts gives

$$z(s) = t + \left[\frac{2x - t^2}{2(t + 1)} \right].$$

□

2. WAVE EQUATION

2.1. 1-Dimensional Variant. We consider the wave equation:

$$u_{tt} = c^2 u_{xx}, \quad c > 0 \quad (2.1)$$

where $u \in C^2(\mathbb{R})$ is our unknown function. The associated initial value problem goes as

$$\begin{cases} u_{tt} = c^2 u_{xx}, & x \in \mathbb{R}, t > 0, \\ u(x, 0) = \phi(x), & x \in \mathbb{R}, \\ u_t(x, 0) = \psi(x), & x \in \mathbb{R}. \end{cases} \quad (2.2)$$

THEOREM 2.1. *Let u be a solution to (2.1). Then*

(1) *There exist functions f and g such that*

$$u(x, t) = f(x + ct) + g(x - ct).$$

(2) *If u satisfies the IVP (2.2) then*

$$u(x, t) = \frac{\phi(x + ct) - \phi(x - ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(\tau) d\tau.$$

The formula above continues to hold (to a certain degree) when we introduce a “source term” to the IVP (2.2). Consider the IVP

$$\begin{cases} u_{tt} - c^2 u_{xx} \equiv f(x, t), & x \in \mathbb{R}, t > 0, \\ u(x, 0) = \phi(x), & x \in \mathbb{R}, \\ u_t(x, 0) = \psi(x), & x \in \mathbb{R}. \end{cases} \quad (2.3)$$

Then we have the following analogue of Theorem 2.1.

THEOREM 2.2. *If u solves (2.3) then*

$$u(x, t) = \frac{\phi(x + ct) - \phi(x - ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(\tau) d\tau + \frac{1}{2c} \int_{\mathfrak{D}} f(y, s) dA.$$

Here $\mathfrak{D}$ is the domain of dependence associated with (x, t) .

This also extends to problems involving *semi-infinite* strings. These are fixed at one end (traditionally, at $x = 0$). The corresponding problem for a semi-infinite string is given below

$$\begin{cases} v_{tt} \equiv c^2 v_{xx}, & x \geq 0, t > 0, \\ v(x, 0) = \phi(x), & x \geq 0, \\ v_t(x, 0) = \psi(x), & x \geq 0, \\ v(0, t) = 0, & t \geq 0. \end{cases} \quad (2.4)$$

THEOREM 2.3. *If v satisfies (2.4) then*

$$u(x, t) = \frac{1}{2} \begin{cases} \phi(x + ct) + \phi(x - ct) + \frac{1}{c} \int_{x-ct}^{x+ct} \psi(\tau) d\tau, & x \geq ct, \\ \phi(x + ct) - \phi(ct - x) + \frac{1}{c} \int_{ct-x}^{x+ct} \psi(\tau) d\tau, & 0 < x \leq ct. \end{cases}$$

2.2. 3-Dimensional Variant. The wave equation admits a generalization in higher dimensions. We shall denote by Δ the Laplacian in $\mathbb{R}^n$, i.e.

$$\Delta := \sum_{j=1}^n \frac{\partial^2}{\partial x_j^2}.$$

The wave equation in $\mathbb{R}^n$ is simply

$$u_{tt} = c^2 \Delta u, \quad u \in C^2(\mathbb{R}^n).$$

For what remains of this subsection, we assume $n = 3$. The corresponding IVP is

$$\begin{cases} u_{tt} \equiv c^2 \Delta u, & x \in \mathbb{R}^3, t > 0, \\ u(x, 0) = \phi(x), & x \in \mathbb{R}^3, \\ u_t(x, 0) = \psi(x), & x \in \mathbb{R}^3. \end{cases} \quad (2.5)$$

THEOREM 2.4 (Kirchoff). *Let u solve (2.5). Then*

$$u(x, t) = \frac{1}{4\pi c^2 t^2} \iint_{\partial B(x, ct)} [\phi(y) + \nabla \phi(y) \cdot (y - x) + t\psi(y)] dS(y).$$

3. DISTRIBUTIONS & FOURIER TRANSFORMS

Let $C_c^\infty(\mathbb{R})$ denote the space of all functions $f : \mathbb{R} \rightarrow \mathbb{C}$ that are infinitely differentiable and of compact support. Note that $C_c^\infty(\mathbb{R}) \subsetneq \mathcal{S}$, where $\mathcal{S}$ is the Schwartz space. One can develop the theory that follows using $\mathcal{S}$ instead of $C_c^\infty(\mathbb{R})$, in which case one obtains the theory of “tempered distributions”.

We topologize $C_c^\infty(\mathbb{R})$ by attaching the sup-norm $\|\cdot\|_\infty$; we will then say that a sequence $\{\phi_k\}_{k=1}^\infty$ in $C_c^\infty(\mathbb{R})$ converges to $\phi \in C_c^\infty(\mathbb{R})$ provided

$$\|\partial^\alpha \phi_k - \partial^\alpha \phi\|_\infty \xrightarrow{k \rightarrow \infty} 0$$

for every multi-index α . Under this notion of convergence, we shall write $\mathcal{D}(\mathbb{R}^n)$ to mean $C_c^\infty(\mathbb{R}^n)$.

DEFINITION 1. Let $f \in L_{\text{loc}}^1(\mathbb{R}^n)$. The distribution induced by f , denoted F_f , is the map

$$\langle F_f, \phi \rangle := \int_{\mathbb{R}^n} f(x) \phi(x) dx.$$

DEFINITION 2. If F is a distribution and α is a multi-index then

$$\langle \partial^\alpha F, \phi \rangle := (-1)^{|\alpha|} \langle F, \partial^\alpha \phi \rangle.$$

DEFINITION 3. Let $\{F_n\}_n$ be a sequence of distributions and F a distribution. We say $F_n \rightarrow F$ in distribution if for all $\phi \in \mathcal{D}(\mathbb{R}^n)$ there holds

$$\lim_{n \rightarrow \infty} \langle F_n, \phi \rangle = \langle F, \phi \rangle.$$

PROPOSITION 3.1. Let F_n be a sequence of distributions converging to F in distribution. Then $\partial^\alpha F_n$ converges to $\partial^\alpha F$ in distribution for every multi-index α .

Proof. Let $\phi \in \mathcal{D}(\mathbb{R}^n)$;

$$\lim_{n \rightarrow \infty} \langle \partial^\alpha F_n, \phi \rangle = (-1)^{|\alpha|} \lim_{n \rightarrow \infty} \langle F_n, \partial^\alpha \phi \rangle = (-1)^{|\alpha|} \langle F, \partial^\alpha \phi \rangle = \langle \partial^\alpha F, \phi \rangle.$$

□

If one knows the theory of abstract measure spaces and integration the theory of distributions is more or less complete after this point.

3.1. Fourier Transforms. We very quickly introduce the idea of a Fourier transform in $\mathbb{R}^n$. On my [website](#) one can find a rigorous treatment of this for functions in $\mathcal{S}$, the Schwartz space of rapidly decreasing functions.

DEFINITION 4. Let $f \in L^1(\mathbb{R}^n)$. The Fourier transform of f , denoted $\hat{f}$, is the function defined by

$$\hat{f}(k) := \int_{\mathbb{R}^n} f(x) e^{-ik \cdot x} dx.$$

Since $f \in L^1(\mathbb{R}^n)$ and $|e^{-ik \cdot x}| \leq 1$, this is well defined.

As on $\mathbb{R}$, several nice properties continue to hold provided we make some integrability assumptions on $\hat{f}$. We point out that $L^2(\mathbb{R}^n)$ is really the “right” setting for Fourier analysis, but we shall bypass this for the sake of readers who have not seen any functional analysis.

THEOREM 3.2 (Fourier Inversion Formula). Let $f \in L^1(\mathbb{R}^n)$ be such that $\hat{f} \in L^1(\mathbb{R}^n)$. Then one can recover f from $\hat{f}$ as follows:

$$f(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(k) e^{ik \cdot x} dk.$$

We then adopt the notation $f = [\hat{f}]^\vee$.

The Fourier transform continues to satisfy many nice properties. We list these below, without proof.

Translation: If $a \in \mathbb{R}^n$ and $f \in L^1(\mathbb{R}^n)$ then

$$\widehat{f(x-a)} = e^{-ik \cdot a} \hat{f}(k) \quad \text{and} \quad \widehat{[e^{-ia \cdot x} f(x)]} = \hat{f}(k-a).$$

Scaling: For a function $g : \mathbb{R}^n \rightarrow \mathbb{C}$ and $c > 0$ define

$$g_c(y) := \frac{1}{c^n} f(y/c).$$

Then, if $f \in L^1(\mathbb{R}^n)$:

$$\widehat{f(cx)} = \left[\widehat{f} \right]_c(k) \quad \text{and} \quad \widehat{f_c}(k) = \widehat{f}(ck).$$

Separation: If f is a function in vectors of the form $(x_1, x_2, \dots, x_n)$ and

$$f(x) = \prod_{j=1}^n f_j(x_j),$$

for some functions $f_j \in L^1(\mathbb{R})$, then $\widehat{f}(k) = \prod_{j=1}^n \widehat{f_j}(k)$.

4. HARMONIC FUNCTIONS

In this section we consider harmonic functions, mostly on $\mathbb{R}^3$. Considering these functions in 2 or 3 dimensions will simplify the formulas. However, we point out that the qualitative results hold in any space dimension. Unless stated otherwise, $\Omega \subset \mathbb{R}^n$ is a bounded and connected open set (these are called domains).

DEFINITION 5. A function $u \in C^2(\Omega) \cap C(\overline{\Omega})$ is called **harmonic** provided $\Delta u \equiv 0$ in Ω . Given a function g that is continuous on $\partial\Omega$, the associated Dirichlet problem is

$$\begin{cases} \Delta u \equiv 0, & \text{in } \Omega, \\ u \equiv g, & \text{on } \partial\Omega. \end{cases} \quad (4.1)$$

The results that follow are beautiful properties of harmonic functions. Many of these continue to hold even for general elliptic PDE.

THEOREM 4.1 (Mean Value Property). *Let $u \in C^2(\Omega)$ be harmonic in Ω and suppose $B(x_0, r)$ is an open ball contained, along with its boundary, in Ω . Then,*

$$u(x_0) = \oint_{B(x_0, a)} u(x) \, dx = \oint_{\partial B(x_0, a)} u(x) \, dS.$$

Here ‘ $\oint$ ’ denotes the mean integral with respect to the Lebesgue measure.

THEOREM 4.2 (Strong Maximum Principle). *Let $u \in C^2(\Omega) \cap C(\overline{\Omega})$ be harmonic in Ω . If u achieves a minimum or maximum in the interior of Ω , then u is constant on $\overline{\Omega}$.*

COROLLARY 4.3 (Weak Maximum Principle). *Let $u \in C^2(\Omega) \cap C(\overline{\Omega})$ be harmonic in Ω . Then*

$$\max_{\overline{\Omega}} u(x) = \max_{\partial\Omega} u(x).$$

COROLLARY 4.4. *System (4.1) admits at most one $C^2(\Omega) \cap C(\overline{\Omega})$ solution.*

4.1. Green's Functions for Harmonic Functions. In this subsection we consider the Laplace equation $\Delta u \equiv 0$ in $\mathbb{R}^3$, i.e. with $n = 3$. For the remainder of this section, Ω will denote a domain in $\mathbb{R}^3$.

One can verify by direct calculation that the function

$$\Phi(x) = -\frac{1}{4\pi|x|}$$

satisfies $\Delta\Phi(x) = \delta_0$ in the sense of distributions. Hence, $\Delta\Phi(x - x_0) = \delta_{x_0}$, in the distributional sense. This function Φ will be called the *fundamental solution* of the Laplacian.

THEOREM 4.5. *Given a function $f \in C_c^\infty(\mathbb{R}^3)$ then*

$$u(x) := \iiint_{\mathbb{R}^3} \Phi(x - y)f(y) \, dy$$

solves $\Delta u \equiv f$.

We also have the following representation formula for functions harmonic in domains $\Omega \subset \mathbb{R}^3$.

THEOREM 4.6. *Let $u \in C^2(\overline{\Omega})$ be harmonic in Ω . For each $x_0 \in \Omega$:*

$$u(x_0) = \iint_{\partial\Omega} \left[u \frac{\partial\Phi(x - x_0)}{\partial\mathbf{n}} - \Phi(x - x_0) \frac{\partial u}{\partial\mathbf{n}} \right] \, ds.$$

We now turn towards obtaining solutions to (4.1) for nice enough g . This involves introducing the notion of a Green's function.

DEFINITION 6. Let $x_0 \in \Omega$. The Green's function for Δ on Ω with source at x_0 , denoted $G(x, x_0)$, is a function defined on $\Omega \setminus \{x_0\}$ such that

- (1) The function $H(x)$ defined by

$$H(x) := G(x, x_0) - \Phi(x - x_0) = G(x, x_0) + \frac{1}{4\pi|x - x_0|}$$

is smooth and harmonic in Ω .

- (2) $G(x, x_0) = 0$ for all $x \in \partial\Omega$.

4.1.1. Algorithm for Green's Function. The typical algorithm for finding a Green's function for a domain Ω goes as follows (we fix $x_0 \in \Omega$)

- (i) Find a harmonic smooth function $H(\mathbf{x})$ equal to $-\Phi(\mathbf{x} - \mathbf{x}_0)$ for $x \in \partial\Omega$.
- (ii) Take $G(\mathbf{x}, \mathbf{x}_0) := H(\mathbf{x}) + \Phi(\mathbf{x} - \mathbf{x}_0)$.

PROPOSITION 4.7. *For all $x \neq x_0$ one has $G(x, x_0) = G(x_0, x)$.*

5. FOURIER SERIES

There is significant preparation in the theory of Fourier series. Consider a function ϕ defined on the interval $(0, l)$, for some $l \in (0, \infty)$. Our first goal is to determine coefficients b_n such that

$$\phi(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right).$$

We give the following definition in preparation of our results;

DEFINITION 7. Let ϕ be defined on $(-l, l)$.

(1) The Fourier sine series for ϕ on $(0, l)$ is

$$\sum_{n \geq 1} b_n \sin\left(\frac{n\pi x}{l}\right), \quad b_n := \frac{2}{l} \int_0^l \phi(x) \sin\left(\frac{n\pi x}{l}\right) dx, \quad n \geq 1.$$

(2) The Fourier cosine series for ϕ on $(0, l)$ is

$$\frac{a_0}{2} + \sum_{n \geq 1} a_n \cos\left(\frac{n\pi x}{l}\right), \quad a_n := \frac{2}{l} \int_0^l \phi(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad n \geq 0.$$

(3) The full Fourier series for ϕ on $(-l, l)$ is given by

$$\frac{a_0}{2} + \sum_{n \geq 1} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

where

$$a_n := \frac{1}{l} \int_{-l}^l \phi(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad n \geq 0,$$

and

$$b_n := \frac{1}{l} \int_{-l}^l \phi(x) \sin\left(\frac{n\pi x}{l}\right) dx, \quad n \geq 1.$$

We comment on the subtleties of these definitions.

- (1) The Fourier sine series is defined on $\mathbb{R}$ and is an odd function of period $2l$.
- (2) The Fourier cosine series is defined on $\mathbb{R}$ and is an even function of period $2l$.
- (3) The full Fourier series is defined for $x \in \mathbb{R}$ and has period $2l$.

We will now explore properties of Fourier series for smooth functions.

5.1. Properties of Fourier Series.

PROPOSITION 5.1. Let ϕ be a sufficiently smooth function on $\mathbb{R}$.

- (1) The Fourier sine series for ϕ on $(0, l)$ will converge to the odd periodic extension of ϕ . The values at integer multiples of l will depend on how ϕ behaves at $\pm l$.
- (2) The Fourier cosine series for ϕ on $(0, l)$ will converge to the even periodic extension of ϕ . The values at integer multiples of l will depend on how ϕ behaves at $\pm l$.
- (3) The full Fourier series for ϕ on $(-l, l)$ will converge to the periodic extension of ϕ . The values at $(2k + 1)l$, with $k \in \mathbb{Z}$, will depend on how ϕ behaves at $\pm l$.

5.1.1. *Eigenfunctions & The Spectrum of ∂^2 .* We will denote by $\langle \cdot, \cdot \rangle$ the $L^2(\mathbb{R})$ inner-product, which makes $L^2(\mathbb{R})$ into a Hilbert space $\mathcal{H}$. Let us denote by L the linear operator on the vector space $C^\infty(\mathbb{R})$ over $\mathbb{R}$.

$$L := -\frac{d^2}{dx^2}.$$

We consider the Eigenvalue problem

$$\begin{cases} X'' + \lambda X \equiv 0, & x \in (0, l), \\ X(0) = X(l) = 0. \end{cases}$$

If $\lambda < 0$ then $L(X) = -\lambda X$ which has general solution

$$X(x) = Ae^{\sqrt{-\lambda}x} + Be^{-\sqrt{-\lambda}x}.$$

However, the boundary conditions give $A = B = 0$ which gives $X \equiv 0$. If $\lambda = 0$ then $X(x)$ is a linear function $Ax + B$. The boundary conditions give $A = B = 0$ which again gives only the trivial solution. If $\lambda > 0$ then the general solution will be of the form

$$X(x) = A \sin(\sqrt{\lambda}x) + B \cos(\sqrt{\lambda}x).$$

Using that $X(0) = B = 0$, we find that $X(x) = A \sin(\sqrt{\lambda}x)$. To avoid the trivial solution, we impose the condition

$$\sin(\sqrt{\lambda}l) = 0.$$

Thus, the possibilities are

$$\lambda = \frac{n^2}{\pi^2 l^2}, \quad n \in \mathbb{N}.$$

We therefore choose Eigenfunctions

$$\sin\left(\frac{\pi nx}{l}\right).$$

Some Hilbert space theory then gives us the usual Fourier representations!

5.2. Bessel and Parseval. We continue to adopt the notation of the previous subsection.

THEOREM 5.2 (Bessel). *Let $\phi \in L^2((a, b))$ and let $\{X_n\}_{n=1}^\infty$ be an orthogonal (non-zero) family of functions in $L^2((a, b))$. Define*

$$A_n := \frac{\langle \phi, X_n \rangle}{\|X_n\|^2}.$$

Then,

$$\sum_{n \geq 1} A_n^2 \int_a^b |X_n(x)|^2 dx \leq \int_a^b |\phi(x)|^2 dx.$$

THEOREM 5.3 (Parseval). *In addition to the previous theorem, the X_n are the eigenfunctions for the Fourier series and A_n their coefficients. Then*

$$\sum_{n \geq 1} A_n^2 \int_a^b |X_n(x)|^2 dx = \int_a^b |\phi(x)|^2 dx.$$

5.3. Differentiation and Integration.

THEOREM 5.4. Let $\phi \in C([-\pi, \pi])$ with $\phi(\pi) = \phi(-\pi)$ and suppose that ϕ is piecewise C^1 on $(-\pi, \pi)$.

- (1) Let $\{a_n\}_{n \geq 0}$ and $\{b_n\}_{n \geq 1}$ be the coefficients of the full Fourier series for ϕ . Let $\{A_n\}_{n \geq 0}$ and $\{B_n\}_{n \geq 1}$ be those for ϕ' . Then,

$$A_0 = 0, \quad A_n = nb_n, \quad B_n = -na_n \quad \text{for } n \in \mathbb{N}.$$

- (2) Suppose further that ϕ' , extended periodically to $\mathbb{R}$, is piecewise C^1 on any interval over which it is continuous. Then

$$\phi'(x) = \sum_{n \geq 1} [nb_n \cos(nx) - na_n \sin(nx)].$$

We also have the following analogue for integration.

THEOREM 5.5. Let ϕ be a piecewise continuous function on $[-\pi, \pi]$ and extend ϕ to $\mathbb{R}$ (with periodicity 2π). Let $\{a_n\}_{n \geq 0}$ and $\{b_n\}_{n \geq 1}$ be the coefficients of the full Fourier series on $(-\pi, \pi)$. Define

$$\Phi(x) := \int_0^x \phi(y) \, dy.$$

Then Φ is continuous and piecewise C^1 . If x is a continuity point of ϕ , then $\Phi'(x) = \phi(x)$. Furthermore, the full Fourier series for Φ is given by

$$\frac{A_0}{2} + \sum_{n \geq 1} \left[\frac{-b_n}{n} \cos(nx) + \frac{a_n}{n} \sin(nx) \right] \quad \text{where } A_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} \phi(y) \, dy.$$

5.4. Uniform Convergence Theorem. Let ϕ be continuous on $[-\pi, \pi]$ and suppose ϕ is piecewise C^1 on $(-\pi, \pi)$ with $\phi(-\pi) = \phi(\pi)$. Then the full Fourier series converges uniformly and absolutely to ϕ on $[-\pi, \pi]$. In fact, it converges uniformly and absolutely to the periodic extension of ϕ on $\mathbb{R}$.